DIPLOMA ECE / ELECTRONICS - SEMESTER 4 - PROGRAM CORE

# Electronics Measurements & Instrumentation

This is a screen-filling handbook for Course Code 4042. It converts the syllabus into a practical study manual: measuring-instrument characteristics, galvanometer conversion, PMMC, MI and electrodynamometer instruments, analog and digital meters, CRO/DSO measurements, bridges, potentiometers, signal generators, analyzers, Q-meter, transducers and recorders.

| | |
|---|---|
| COURSE CODE | 4042 |
| CREDITS | 4 |
| PERIODS | 60 |
| PATTERN | L:3 T:1 P:0 |

□ □□□□□□□□□□□□ □□□ short note □□□□. □□□ instrument-□□□ principle, construction, formula, measurement procedure, common mistake, exam point, and industry use □□□□□□□□□□□□□ □□□□□□□□□ □□□□□□□□□□□□□□□□□□.

OFFICIAL SYLLABUS CONVERTED TO LEARNING ROUTE

# Course Objectives, Outcomes and Module Map

The course objective is to understand testing and measuring instrument characteristics and to learn oscilloscopes, analyzers, recorders, bridges and transducers. Each outcome below is expanded into lesson blocks, formulas, diagrams and practice tasks.

## Course Objectives

- ✓ Learn characteristics of testing and measuring instruments.
- ✓ Understand oscilloscope, analyzer and recorder features.
- ✓ Choose correct instruments for electrical/electronic measurement.
- ✓ Connect theory to troubleshooting, calibration and quality checking.

## Outcome Table

| CO | Outcome | Hours | Cognitive Level |
|---|---|---|---|
| CO1 | Explain the working principle of various measuring instruments. | 15 | Understanding |
| CO2 | Explain the working principle of CRT and CRO. | 15 | Understanding |
| CO3 | Illustrate bridges, signal generators and analyzers. | 16 | Understanding |
| CO4 | Explain the working of transducers and recorders. | 12 | Understanding |
| Series Test | Internal assessment | 2 | - |

## Module Breakdown

| Module | Sub-outcomes | Main contents |
|---|---|---|
| M1 | M1.01 specifications, M1.02 galvanometer conversion, M1.03 PMMC and MI, M1.04 analog and digital meters. | Static characteristics, errors, galvanometer, ammeter, voltmeter, PMMC, MI, electrodynamometer, analog voltmeter, DVM, DMM, energy meter. |

| | | |
|---|---|---|
| M2 | M2.01 CRT, M2.02 CRO, M2.03 CRO measurement, M2.04 DSO features. | CRT construction, deflection systems, delay line, CRO block diagram, dual trace CRO, voltage/time/frequency/phase measurement, Lissajous patterns, DSO, probes. |
| M3 | M3.01 bridges and potentiometers, M3.02 bridge applications, M3.03 function generator, M3.04 analyzers and Q-meter. | Wheatstone, Kelvin double, Maxwell, Hay, DC and AC potentiometers, signal generator, function generator, wave analyzer, spectrum analyzer, Q-meter. |
| M4 | M4.01 transducer types, M4.02 selection criteria, M4.03 working of transducers, M4.04 recorders. | Thermocouple, thermistor, LVDT, strain gauge, piezoelectric transducer, load cell, X-Y recorder, strip chart recorder. |

MODULE 1 - 15 HOURS

# Measurements and Measuring Instruments

Goal: understand measurement characteristics, instrument errors, galvanometer conversion, PMMC/MI/electrodynamometer instruments, analog meters, DVM, DMM and energy meter.

### 1. What is measurement?

Measurement is the comparison of an unknown quantity with an accepted standard. A measurement system must sense the quantity, convert it into a usable form, display/record it and show the result with acceptable accuracy.

□□□ value meter-□ □□□□□□□□□□□□□□□ □□□ automatically correct □□□□□□. Range, loading effect, calibration, zero error, resolution, probe setting □□□□□□ □□□□□□□ □□□□□ □□□□□□□.

### 2. Static characteristics

- **Accuracy:** closeness to true value.
- **Precision:** repeatability of readings.
- **Resolution:** smallest detectable change.
- **Sensitivity:** output change per unit input change.
- **Expected value:** best estimate from repeated observations.

### 3. Types of errors

**Gross error** comes from careless operation or wrong reading. **Systematic error** repeats due to calibration, environmental or instrumental defect. **Random error** changes unpredictably and is reduced by repeated measurement and averaging.

**4. Instrument loading**

A voltmeter should have high resistance; an ammeter should have low resistance. If instrument resistance disturbs the circuit, the measured value changes. This is important when checking control panel sensors, low-level signals and high impedance electronic circuits.

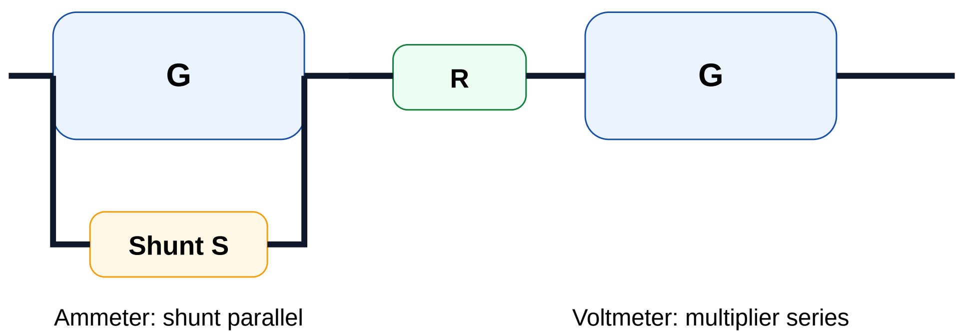
## Galvanometer Conversion

A galvanometer is a sensitive current measuring device. By connecting a low-value shunt resistance in parallel, it becomes an ammeter. By connecting a high-value multiplier resistance in series, it becomes a voltmeter.

Ammeter shunt: $S = Ig\, G / (I - Ig)$

Voltmeter multiplier: $R = V / Ig - G$

Where G is galvanometer resistance, Ig is full scale deflection current, I is required ammeter range and V is required voltmeter range.




Galvanometer conversion principle

## PMMC Instrument

Permanent Magnet Moving Coil instrument uses the interaction between current in a coil and the magnetic field of a permanent magnet. The deflecting torque is proportional to current, so the scale is uniform. It is mainly used for DC measurement. For AC, a rectifier is needed.

- ✓ Uniform scale and high sensitivity.
- ✓ Low power consumption.
- ✓ Polarity sensitive - wrong polarity gives reverse deflection.
- ✓ Not suitable directly for AC because torque reverses with current direction.

Deflecting torque Td = N B I A

PMMC DC-□□□□□□□ best. AC direct □□□□□□□□□□ average torque zero □□□□. □□□□□□□ rectifier type AC meter □□□□.

## Moving Iron Instrument

Moving Iron instruments work on magnetic attraction or repulsion of soft iron pieces. Since deflection depends on $I^2$, it can measure both AC and DC. Scale is non-uniform and crowded near zero.

#### Attraction type

Current in fixed coil magnetizes a soft iron piece and attracts it into the stronger field region.

#### Repulsion type

Two iron pieces are similarly magnetized and repel each other, producing pointer deflection.

- Robust and cheap.
- Suitable for AC/DC current and voltage.
- Lower accuracy than PMMC.
- Errors: hysteresis, frequency, stray field, temperature.

## Electrodynamometer Instrument

It has fixed coils and moving coil. Torque is produced by interaction between fields of fixed and moving coils. It is used in wattmeters and can work on AC and DC.

**Exam point:** electrodynamometer type wattmeter has current coil in series and pressure coil in parallel. It measures true power in AC circuits.

Power in single phase circuit: $P = V I \cos\varphi$

## Analog and Digital Meters

**Analog meter** gives continuous pointer movement; useful for observing trends. **Digital meter** gives numeric display; useful for precision and avoiding parallax error.

**Ramp type DVM**

1. Unknown voltage is compared with a linear ramp voltage.
2. Gate opens when ramp starts and closes when ramp equals input.
3. Counter counts clock pulses during gate time.
4. Count is proportional to input voltage.

### Digital Multimeter and Energy Meter

DMM combines DC voltage, AC voltage, resistance, continuity, diode test and current measurement. In industry, a good DMM is mandatory for panel troubleshooting. Single-phase energy meter measures energy consumption in kWh.

> Electrical energy = Power × Time; 1 kWh = 1000 W for 1 hour

**Panel safety:** Never measure current by placing the meter in parallel. Put ammeter in series or use clamp meter where suitable. Wrong DMM jack selection can damage the meter and circuit.

**MODULE 2 - 15 HOURS**

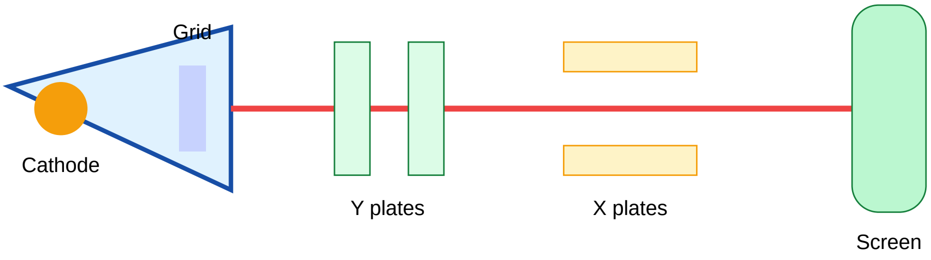
# CRT, CRO, DSO and Measurements

Goal: understand cathode ray tube, CRO block diagram, vertical/horizontal deflection, delay line, dual trace oscilloscope, DSO, probes and measurements using CRO.

### Cathode Ray Tube Construction

CRT converts electrical signals into visible display. Main parts are electron gun, control grid, focusing anode, accelerating anode, vertical plates, horizontal plates and fluorescent screen.

- ✓ Electron gun produces electron beam.
- ✓ Control grid controls brightness.
- ✓ Focusing system makes a sharp spot.
- ✓ Deflection plates move the beam horizontally and vertically.
- ✓ Screen glows where the beam strikes.




Basic CRT section

## CRO Block Diagram

A CRO displays voltage as a function of time. Vertical amplifier controls vertical deflection; time base generator produces sawtooth voltage for horizontal deflection. Trigger circuit synchronizes the waveform.

1. Input signal enters through attenuator and vertical amplifier.
2. Delay line delays vertical signal slightly so the starting edge is visible.
3. Time base generator drives horizontal plates.
4. Trigger locks waveform and prevents rolling.
5. CRT displays waveform on screen graticule.

CRO-□□□ waveform stable □□□□□□□□□□□ trigger correct □□□□. Otherwise trace left/right roll □□□□□□□□.

## Measurements Using CRO

Voltage = vertical divisions × volts/div

Time period T = horizontal divisions × time/div

Frequency f = 1 / T

Phase angle $\theta$ = (horizontal shift / one cycle width) × 360°

For DC voltage, set coupling to DC and count the vertical shift from zero reference. For AC voltage, measure peak-to-peak value and convert as needed.

### Lissajous Pattern Frequency Ratio

When two sine waves are applied to X and Y plates, stationary Lissajous pattern is formed if the frequency ratio is simple.

$f_x / f_y$ = number of vertical tangencies / number of horizontal tangencies

For phase measurement with equal frequencies, an ellipse pattern is used. Phase can be obtained from the intercept method.

### Multiple Trace and Dual Trace CRO

A multiple trace CRO displays more than one signal. A dual trace CRO uses an electronic switch to display two channels on the same CRT. It works in alternate mode for high frequencies and chop mode for low frequencies.

- **Alternate mode:** displays CH1 in one sweep and CH2 in next sweep.
- **Chop mode:** rapidly switches between channels during one sweep.

### Digital Storage Oscilloscope

DSO samples the input waveform using ADC and stores samples in memory. It can display single-shot events, save waveforms, calculate measurements and communicate with computers.

**Block idea:** Input attenuator - vertical amplifier - ADC - memory - processing - display.

**Probe warning:** 10:1 probe reduces loading and increases input impedance, but the scope must be set to 10X. Wrong setting gives wrong displayed voltage.

MODULE 3 - 16 HOURS

## Bridges, Potentiometers, Signal Generators, Analyzers and Q-meter

Goal: use bridge balance principles for resistance, inductance and Q measurement; understand signal generation and waveform/frequency analysis.

### Wheatstone Bridge

Used for medium resistance measurement. At balance, no current flows through galvanometer.

> At balance: $P/Q = R/S$, therefore unknown $S = Q\,R / P$

Accuracy depends on known resistors, galvanometer sensitivity and proper null balance.

### Kelvin Double Bridge

Used for low resistance measurement. It reduces lead and contact resistance error by using additional ratio arms. This is useful for measuring winding resistance, busbar joints and low-ohm conductors.

Low resistance measure □□□□□□□□□□□ test lead resistance □□□□□ □□□□ error □□□□□□□□□□. Kelvin method □□□□□ compensate □□□□□□□□□□□.

### AC Bridges

**Maxwell's inductance bridge** measures low-Q inductors using known resistance and capacitance. **Hay's bridge** is suitable for high-Q inductors.

- AC bridge balance requires magnitude and phase balance.
- Detector may be headphones, vibration galvanometer or electronic detector.
- Supply frequency and standard component quality affect accuracy.

### Potentiometers

A DC potentiometer measures unknown emf by comparison with a standard cell. Since balance condition draws no current from the unknown source, it is very accurate.

1. Standardize the potentiometer using a standard cell.
2. Connect unknown emf.
3. Adjust sliding contact until null is obtained.
4. Calculate emf proportional to balancing length.

### Signal Generator and Function Generator

A signal generator produces test signals of known frequency and amplitude. A function generator can generate sine, square and triangular waveforms. It is used for amplifier testing, filter response, CRO practice and circuit troubleshooting.

- ✓ Frequency range selection.
- ✓ Amplitude adjustment.
- ✓ Waveform selection.
- ✓ Offset control in advanced generators.
- ✓ Output impedance usually 50 ohm in lab instruments.

### Wave Analyzer and Spectrum Analyzer

A wave analyzer measures amplitude of individual frequency components. A spectrum analyzer displays frequency content of a signal over a range, showing fundamental, harmonics and noise.

**Practical use:** finding harmonics, checking noise in power supply outputs, RF troubleshooting and vibration/electrical signature analysis.

### Q-meter

Q-meter measures quality factor of a coil using resonance principle. At resonance, inductive reactance equals capacitive reactance and voltage across capacitor is Q times the applied voltage in a series resonant circuit.

$$Q = X_L / R = 2\pi f L / R$$

$$\text{At resonance: } f_r = 1 / (2\pi\sqrt{LC})$$

High Q means lower losses and sharper resonance. Q-meter applications include measuring Q, inductance, distributed capacitance and dielectric loss.

**MODULE 4 - 12 HOURS**

## Transducers and Recorders

Goal: classify transducers, select suitable transducer for application, understand thermocouple, thermistor, LVDT, strain gauge, piezoelectric transducer, load cell, X-Y recorder and strip chart recorder.

## Transducer Basics

A transducer converts a physical quantity into an electrical signal. A sensor detects the physical condition, while a transducer provides usable electrical output.

| Classification | Examples |
|---|---|
| Active | Thermocouple, piezoelectric transducer |
| Passive | Thermistor, LVDT, strain gauge |
| Analog | LVDT voltage, thermistor resistance |
| Digital | Encoder, digital temperature sensor |

## Selection Criteria

- ✓ Range and sensitivity.
- ✓ Accuracy and linearity.
- ✓ Response time.
- ✓ Environmental suitability: heat, moisture, vibration.
- ✓ Mechanical mounting and size.
- ✓ Output type and signal conditioning needed.
- ✓ Cost, reliability and maintenance.

Escalator/control panel work-□ sensor □□□□□□□□□□□□□□□□□□ range □□□□□□ □□□□□□□□□. Noise, cable length, mounting, vibration, maintenance □□□□□□□□□ □□□□□□□.

## Temperature Transducers

**Thermocouple:** works on Seebeck effect. Two dissimilar metals produce emf proportional to temperature difference.

**Thermistor:** semiconductor resistor with high temperature coefficient. NTC thermistor resistance decreases as temperature increases.

> Thermocouple output is small, so amplification and cold-junction compensation are required.

## LVDT

Linear Variable Differential Transformer measures linear displacement. It has one primary and two secondary coils. A movable core changes coupling. Output voltage magnitude and phase indicate displacement and direction.

- Frictionless operation.
- High resolution.
- Requires AC excitation.
- Used in displacement, position and vibration measurement.

## Strain Gauge and Load Cell

A strain gauge changes resistance due to mechanical strain. It is usually used in a Wheatstone bridge. A load cell uses strain gauges to convert force or weight into electrical signal.

> Gauge factor $GF = (\Delta R/R) / \text{strain}$

Industrial weighing systems, force measurement and mechanical stress monitoring use load cells.

## Piezoelectric Transducer

Piezoelectric materials generate electric charge when mechanical stress is applied. They are good for dynamic pressure, vibration and acceleration measurement, but not suitable for static measurement because charge leakage occurs.

## Recorders

A recorder stores variation of a quantity with time or another variable. **Strip chart recorder** records signal versus time. **X-Y recorder** plots one electrical quantity against another, useful for characteristics and calibration curves.

1. Input signal is conditioned and amplified.
2. Servo mechanism drives pen or marker.
3. Chart paper or digital memory stores trace.
4. Recorded data is used for analysis, proof and fault history.

QUICK REVISION

# Formula Bank and Exam Memory Sheet

Use this section before tests. Do not memorize blindly; understand where each formula is applied.

## Measurement Error

Absolute error = measured value - true value

Relative error = absolute error / true value

Percentage error = relative error × 100

## Galvanometer

Shunt $S = I_g G / (I - I_g)$

Multiplier $R = V/I_g - G$

## CRO

V = divisions × volts/div

T = divisions × time/div

$f = 1/T$

## Bridge

Wheatstone balance: $P/Q = R/S$

Unknown $S = QR/P$

### Resonance and Q

$$f_r = 1/(2\pi\sqrt{LC})$$

$$Q = X_L/R = 2\pi f L/R$$

### Transducer

Sensitivity = output change/input change

Gauge factor = $(\Delta R/R)$/strain

WORKED EXAMPLES

## Solved Problems

These are exam-style examples for calculations and reasoning.

---

#### Example 1 - Galvanometer as Ammeter

A galvanometer has resistance 100 ohm and full-scale current 1 mA. Convert it into a 1 A ammeter.

$$S = I_g G/(I - I_g) = (0.001 \times 100)/(1 - 0.001) = 0.100/0.999 = 0.1001 \text{ ohm}$$

**Answer:** Connect approximately 0.1 ohm shunt in parallel.

#### Example 2 - CRO Frequency Measurement

One cycle occupies 5 divisions. Time/div = 0.2 ms/div. Find frequency.

$$T = 5 \times 0.2 \text{ ms} = 1 \text{ ms}; f = 1/T = 1/0.001 = 1000 \text{ Hz}$$

**Answer:** 1 kHz.

### Example 3 - Wheatstone Bridge

In a Wheatstone bridge, P = 100 ohm, Q = 200 ohm and R = 50 ohm. Find unknown S.

$$S = QR/P = (200 \times 50)/100 = 100 \text{ ohm}$$

### Example 4 - Q of Coil

A coil has inductance 20 mH and resistance 4 ohm at 1 kHz. Find Q.

$$Q = 2\pi fL/R = 2\pi \times 1000 \times 0.02 / 4 = 31.4$$

PRACTICE BANK

# Important Questions

Use these questions after reading each module. Write answers with diagram, principle, construction, working, formula and application where required.

## Short Answer

1 Define accuracy, precision and resolution.

2 Differentiate gross, systematic and random errors.

3 Why should a voltmeter have high resistance?

4 State the principle of PMMC instrument.

5 Why is MI instrument suitable for AC and DC?

6 What is the need of a delay line in CRO?

7 Write the formula for frequency measurement using CRO.

8 What is a Lissajous pattern?

9 State Wheatstone bridge balance condition.

10 Why is Kelvin double bridge used?

11 Define Q factor.

12 List active and passive transducers.

13 What is cold junction compensation?

14 Why is piezoelectric transducer not ideal for static measurement?

15 List applications of strip chart recorder.

### Long Answer

1 Explain construction and working of PMMC instrument with torque equation.

2 Describe conversion of galvanometer into ammeter and voltmeter with derivation.

3 Explain CRO block diagram and function of each block.

4 Explain voltage, time, frequency and phase measurement using CRO.

5 Explain DSO block diagram and advantages over analog CRO.

6 Describe Wheatstone bridge and Kelvin double bridge with applications.

7 Explain Maxwell and Hay bridges and their use.

8 Explain function generator and spectrum analyzer.

9 Explain working of LVDT with diagram.

10 Explain thermocouple, thermistor, strain gauge and load cell.

**MODEL EXAMINATION**

# Sample Question Paper - Official Pattern Practice

This is a practice paper based on the syllabus structure. It is not copied from any official question paper.

Course Code: 4042
Course Title: Electronics Measurements & Instrumentation
Time: 3 Hours Maximum Marks: 100

PART A - Answer all questions. Each carries 3 marks.
1. Define sensitivity and resolution of an instrument.
2. What are random errors? How can they be reduced?
3. State the principle of PMMC instrument.
4. Why is a shunt connected in parallel with a galvanometer?
5. List the main parts of CRT.
6. What is the function of time base generator in CRO?
7. State Wheatstone bridge balance condition.
8. What is a function generator?
9. List any four transducer selection criteria.
10. Define strip chart recorder.

PART B - Answer any five questions. Each carries 6 marks.
11. Explain accuracy, precision and expected value with example.
12. Derive the expression for shunt resistance used to convert a galvanometer into an ammeter.
13. Compare PMMC and MI instruments.
14. Explain voltage and frequency measurement using CRO.
15. Explain dual trace CRO with alternate and chop modes.
16. Explain Kelvin double bridge and its application.
17. Explain Q-meter principle and applications.
18. Explain thermocouple and thermistor with applications.

PART C - Answer any five questions. Each carries 8 marks.
19. Explain construction and working of CRT with neat diagram.
20. Explain CRO block diagram and function of each section.
21. Explain Wheatstone bridge and solve a resistance measurement problem.
22. Explain Maxwell bridge and Hay bridge with applications.
23. Explain spectrum analyzer and wave analyzer.
24. Explain LVDT construction, working and characteristics.
25. Explain strain gauge and load cell measurement system.
26. Explain X-Y recorder and strip chart recorder with block diagram.

ANSWER KEY

# Expected Answer Points

### Part A answer points

1. Sensitivity is output change per unit input; resolution is smallest input change that can be detected.
2. Random errors are unpredictable variations; reduce by repeated readings and averaging.
3. PMMC works on force on current carrying coil in magnetic field.
4. Shunt bypasses excess current and protects galvanometer.
5. Electron gun, control grid, focusing/accelerating anodes, deflection plates, screen.
6. Time base generator produces sweep voltage for horizontal deflection.
7. P/Q = R/S at balance.
8. Instrument producing sine, square and triangular waveforms.
9. Range, accuracy, sensitivity, linearity, response time, environment, cost.
10. Recorder that plots signal variation versus time on moving chart.

### Long answer structure

For every 6/8 mark answer: write definition, principle, neat diagram/block diagram, construction, working, important formula, advantages/limitations and applications. For CRO questions, include controls and measurement procedure. For transducer questions, include input quantity, output quantity, signal conditioning and practical use.

REFERENCES

## Books and Online Resources


| Type | Reference |
|---|---|
| T1 | Electronic Instrumentation - H S Kalsi |
| T2 | A Course in Electrical and Electronic Measurements and Instrumentation - A K Sawhney |
| T3 | Electronics Measurements And Instrumentation - U.A Bakshi and A.V. Bakshi |
| R1 | Electronic Instrument and Measurement Technique - W.D Cooper |
| R2 | Electronic Measurement and Instrumentation - J.G Joshi |
| R3 | Electronics and Electrical Measurements and Instrumentation - J.B Gupta |
| R4 | Industrial Electronics and Control - Biswanath Paul |